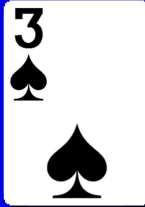
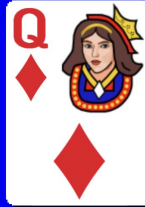
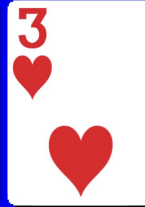
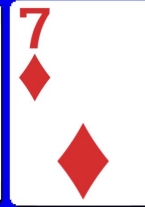
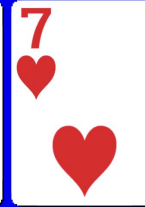


Statement of Purpose / Proposed Specification

A playable VGA-based video poker game that uses a random deck and keeps track of balance over multiple games

- Five-card draw video poker with a starting balance of 100 credits
- Fixed 5-credit wager per round with payouts from 5–4000 credits based on hand
- Player uses switches to hold cards and debounced buttons to deal/draw
- VGA output displays cards, held cards, balance, and round results
- VGA driven by a 25 MHz clock
- PRNG generates cards; seed is sampled from bidirectional I/O switches on asynchronous reset
- Seed switches are available for development/testing, but not exposed to players in production
- Asynchronous reset restores the initial game state and balance

ROYAL FLUSH.....	250	500	750	1000	4000
STRAIGHT FLUSH.....	50	100	150	200	250
FOUR OF A KIND.....	25	50	75	100	125
FULL HOUSE.....	9	18	27	36	45
FLUSH.....	6	12	18	24	30
STRAIGHT.....	4	8	12	16	20
THREE OF A KIND.....	3	6	9	12	15
TWO PAIR.....	2	4	6	8	10
JACKS OR BETTER.....	1	2	3	4	5

HOLD		TWO PAIR		HOLD		HOLD	
3	Q	3	7	7			
							

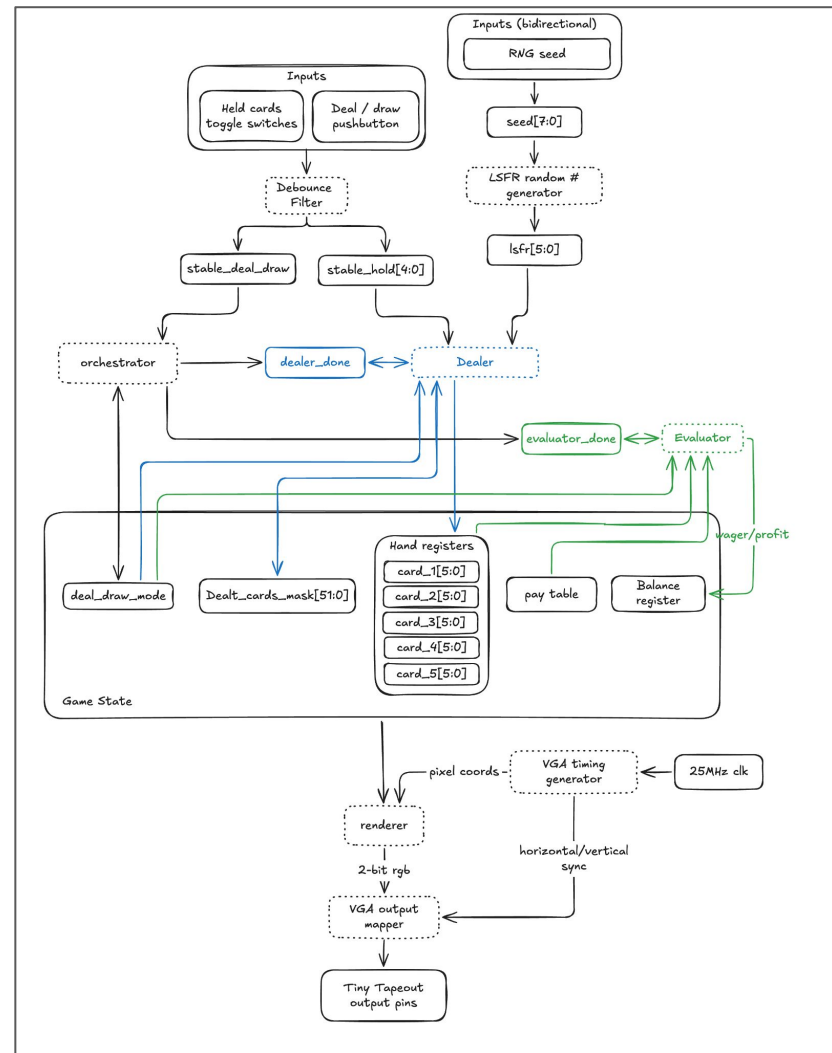
BET 5 9/6 JACKS OR BETTER **CREDITS 1220**

TRAINER OFF MORE GAMES VIEW LOG 2X PAY SOUND MID DRAW

An example of a video poker game (<https://www.freeslots.com/poker.htm>)

System Diagram / IO Pin Assignments

Tiny tapeout pin	Assignment
uo_out[0:7]	VGA
clk	25 MHz system and VGA pixel clock
rst_n	Asynchronous system reset
ena	Tiny Tapeout project enable
ui_in[0:4]	Hold cards 1-5
ui_in[5]	deal/draw pushbutton
ui_in[6:7]	-
uio_in[7:0]	RNG seed
uio_out[7:0]	Driven zero
uio_oe[7:0]	Driven zero



Timeline for Completion

Sept. 24 — Proposal

- Finalize architecture, specifications, system diagram, FSM, and I/O assignments.

Oct. 4 — Core modules

- Implement VGA timing, LFSR, card registers, unique-card dealer, and initial tests.

Oct. 8 — Verilog design and test

- Integrate hold/draw logic, hand evaluator, game FSM, and basic VGA rendering.
- Complete directed Cocotb tests.

Oct. 12–22 — System integration

- Add input filtering, payouts, balance tracking, card graphics, and randomized tests.

Nov. 5 — Design evaluation

- Complete RTL verification, synthesis, utilization analysis, and timing checks.

Nov. 12–19 — Physical implementation

- Perform placement, routing, parasitic extraction, and gate-level verification.

Nov. 26 — System verification

- Resolve remaining issues and ensure all GitHub Actions pass.

Dec. 3 — Final submission

- Complete documentation and submit the Tiny Tapeout design.

Who does what

Nash — Game Logic

- Game state machine & round flow
- Game orchestration / control logic
- Card dealing & deck management
- Poker hand evaluation & payouts

Luke — I/O & Hardware Interface

- Player input handling
- Button debouncing
- Pseudorandom number generator & seeding
- VGA controller & game display